Smart Swarm Robotic System for Earthquake Search and Rescue



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Project Framework and Overview



- Earthquake-damaged environments can be unknown and difficult to search systematically.
- Two simulated robots cooperate to explore, map, share information, and plan a route to a virtual target.



Unknown Environment → **LiDAR Sensing** → **Occupancy Grid Mapping** → **Cooperative Exploration** → **Map and Data Sharing** → **Virtual Target Localization** → **Rescue-Route Planning**

This work studies coordinated exploration and route planning in an initially unknown two-dimensional damaged environment, with practical relevance to structured search support.

Literature Review: Three Selected Papers



Jadeja et al. (2024)

‘Survivor Detection Approach for Post Earthquake Search and Rescue Missions Based on Deep Learning Inspired Algorithms’

Deep-learning survivor detection; Faster R-CNN, SSD, YOLOv8, and YOLOv10; provides detection context, while this project uses a predefined virtual target.



Pongsirijinda et al. (2024)

‘Distributed Multi-Robot Potential-Field-Based Exploration with Submap-Based Mapping and Noise-Augmented Strategy’

Distributed exploration using potential fields and submap-based collaborative mapping; informs cooperative exploration and map sharing.



Calzolari et al. (2024)

‘Reinforcement Learning Driven Multi-Robot Exploration via Explicit Communication and Density-Based Frontier Search’

Multi-robot exploration using explicit communication, reinforcement learning, and density-based frontier search; informs coordination and information exchange.

The proposed project focuses on simulated LiDAR, occupancy grids, frontier exploration, and communication.

Comparative Table of Previous Studies



Research Paper	Main Techniques	Strengths	Limitations
Jadeja et al. (2024)	Faster R-CNN, SSD, YOLOv8, YOLOv10	Addresses deep-learning survivor detection	Focuses on detection; this project does not claim visual survivor detection
Pongsirijinda et al. (2024)	Distributed exploration, potential fields, submap mapping, noise augmentation	Connects multi-robot exploration with collaborative mapping	Described focus differs from this project's frontier-based and greedy approach
Calzolari et al. (2024)	Explicit communication, reinforcement learning, density-based frontier search	Connects exploration and communication	Described focus differs from this project's planned methods

Main and Secondary Project Objectives

Main Objective

To design and simulate a cooperative two-robot system for exploration, mapping, and rescue-route planning in a simulated earthquake-damaged environment.

Secondary Objectives

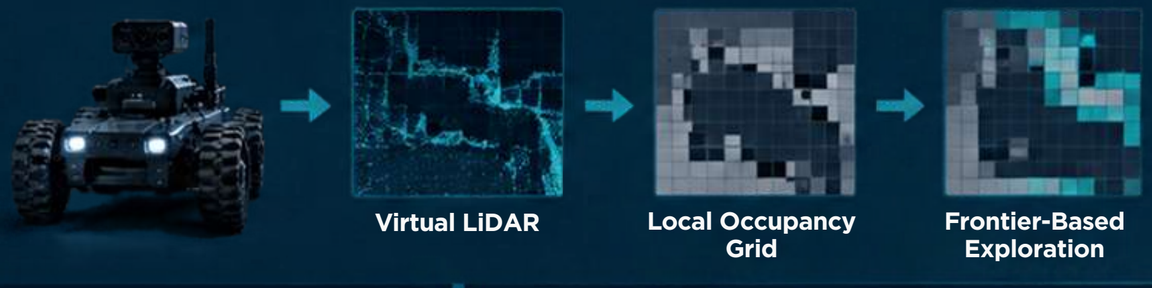
1. Develop a two-dimensional simulated disaster environment.
2. Implement simulated LiDAR sensing and occupancy grid mapping.
3. Apply frontier-based exploration and greedy task allocation.
4. Support communication and information sharing between the robots.
5. Implement A* path planning and replanning.
6. Represent a missing person using a predefined virtual target and a distance threshold.
7. Evaluate exploration coverage, redundant searching, mapping, and rescue-route planning.



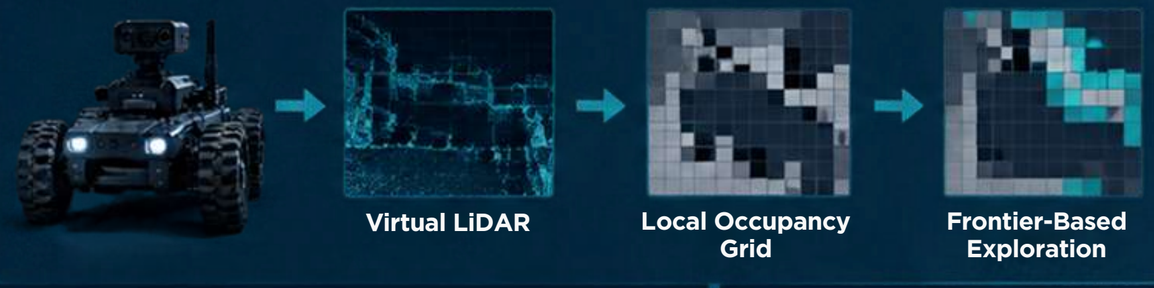
Proposed System Architecture



Robot 1: Virtual LiDAR → Local Occupancy Grid → Frontier-Based Exploration



Robot 2: Virtual LiDAR → Local Occupancy Grid → Frontier-Based Exploration



Communication and Information Exchange

Greedy Task Allocation



Shared Map Integration



Predefined Virtual Target Localization



A* Rescue-Route Planning

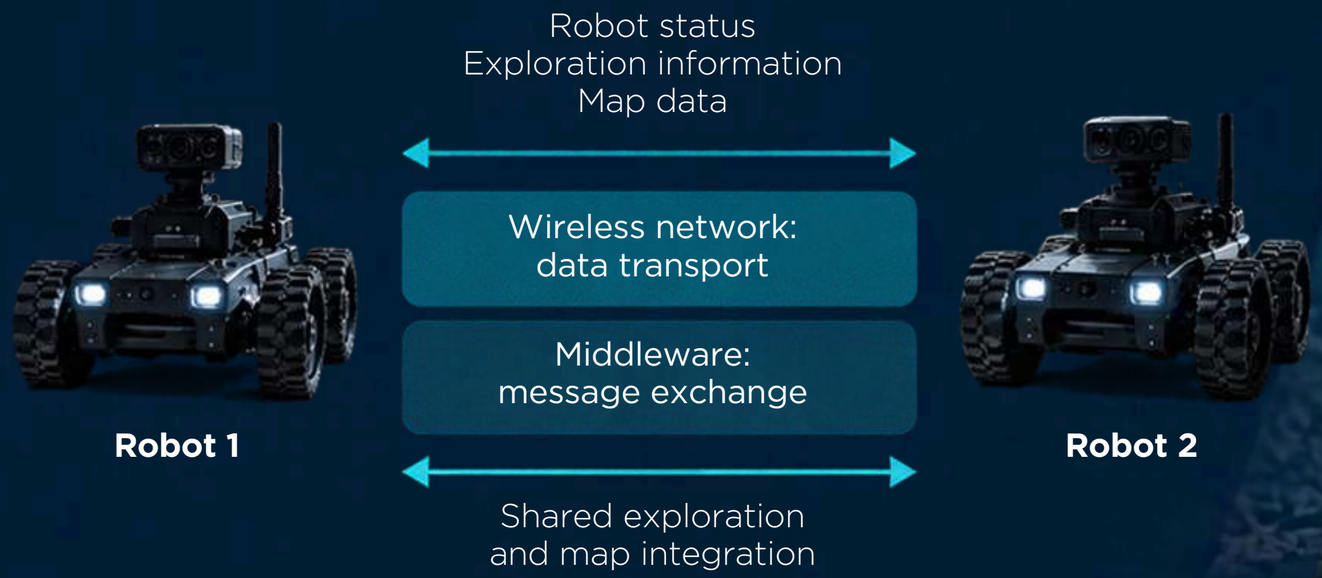


Proposed architecture; implementation status is not asserted.

Technical Specifications

Component	Proposed Specification	Purpose
Robotic Platform	Two cooperative ground robots	Cooperative exploration
Operating Environment	Two-dimensional earthquake-damaged simulation	Search scenario
Sensing System	Simulated LiDAR	Acquire virtual range data
Mapping Method	Occupancy grid mapping	Represent explored and unknown space
Exploration Strategy	Frontier-based exploration	Guide exploration toward unknown boundaries
Task Allocation	Greedy task allocation	Assign exploration tasks
Path Planning	A* algorithm with replanning where required	Plan and update routes
Communication Architecture	ROS 2 Topics and DDS/RTPS, subject to confirmation of the final implementation	Exchange robot information
Information Exchange	Robot status, exploration information, and map data as required by the final design	Support coordination
Map Integration	Combining shared mapping information	Create a shared map representation
Target Representation	Predefined virtual target representing a missing person, detected using a distance threshold	Represent target localization in simulation
Simulation Scope	Software simulation; no physical robot deployment	Define project boundaries
Performance Evaluation	Exploration coverage, redundant exploration, mapping performance, coordination, and rescue-route planning	Assess system behavior

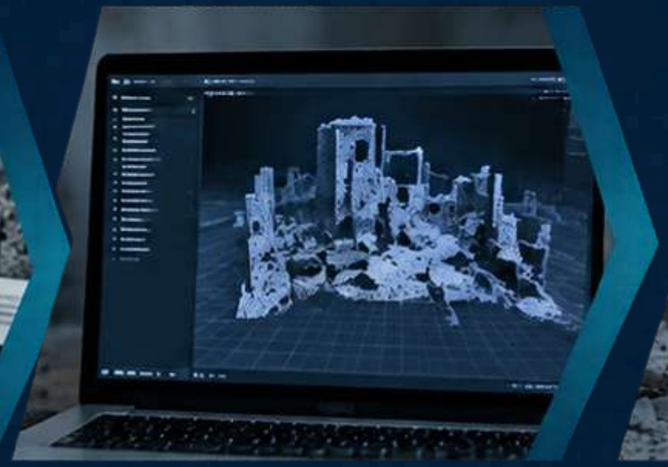
Communication System and Coordination



- Information exchange supports coordinated exploration and reduces redundant searching.
- Robots share status, exploration information, and mapping data as required by the final design.
- Communication is proposed to be implemented using ROS 2 Topics over DDS/RTPS

Adopted Design Methodology









1. Literature Review and Requirements Analysis

establish context, scope, and system requirements.

2. Simulation Environment and System Design

define the simulated environment and proposed system structure.

3. Implementation and Integration

connect mapping, exploration, communication, and path planning.

4. Testing, Performance Evaluation, and Analysis

assess system behavior against the project objectives.

Project Timeline

Provisional schedule — replace editable placeholders with confirmed dates.

- Literature review and requirements analysis
- Simulation environment setup
- Robot modeling and virtual sensing
- Mapping and exploration implementation
- Communication and map-sharing integration
- Task allocation and path planning
- System integration
- Testing and performance evaluation
- Final report and presentation preparation



